

Computer Networking Notes

TCP

- Transmission Control Protocol (TCP) is connection-oriented.
- Ensures reliable, ordered delivery of data.
- Features: 3-way handshake, error detection (checksum), flow control, retransmission.
- Used in applications like HTTP, FTP, SMTP.

UDP

- User Datagram Protocol (UDP) is connectionless.
- Faster but unreliable (no guarantee of delivery/order).
- Used in video streaming, gaming, VoIP.

HTTP

- Hypertext Transfer Protocol, works over TCP/IP.
- Used for communication between client (browser) and server (web).
- Stateless protocol, uses methods like GET, POST, PUT, DELETE.
- Basis of web communication.

Ports

- Ports are numerical identifiers for processes/services.
- Range: 0–65535.
- Examples: 80 (HTTP), 443 (HTTPS), 25 (SMTP), 53 (DNS).

SONET

- Synchronous Optical Network.
- High-speed fiber optic network standard.
- Provides multiplexing, high bandwidth, used in telecom backbone.

Modem & Router

- Modem: Converts digital signals to analog for transmission over telephone lines (and vice versa).
- Router: Directs packets between different networks, forwards based on IP addresses.

OSI Model

- A 7-layer reference model for communication:
 1. Physical
 2. Data Link
 3. Network
 4. Transport
 5. Session
 6. Presentation

7. Application

TCP/IP Model

- 4-layer model:
 1. Link Layer
 2. Internet Layer
 3. Transport Layer
 4. Application Layer

Client-Server Architecture

- Client requests services, server provides them.
- Centralized, scalable but server can be a bottleneck.
- Examples: Web, email, database services.

Peer-to-Peer Architecture

- Each node can act as client and server.
- Decentralized, resource sharing among peers.
- Example: Torrent networks.

HTTP Methods

- GET: Retrieve data.
- POST: Submit data.
- PUT: Update data.
- DELETE: Remove data.
- PATCH: Partial update.

Error/Status Codes

- 1xx: Informational
- 2xx: Success (200 OK)
- 3xx: Redirection (301 Moved Permanently)
- 4xx: Client error (404 Not Found)
- 5xx: Server error (500 Internal Server Error)

Cookies

- Small pieces of data stored by browser.
- Maintains session info, preferences, authentication.
- Key-value format, stored client-side.

Email Working

- Uses SMTP to send mail.
- POP3/IMAP used to receive mail.
- Flow: User → SMTP server → Recipient's server → Recipient fetches via POP3/IMAP.